

Assignment 3

22BCON1068

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B (Information Security System)

• Section - A ($2 \times 2 = 4$)

1. What is difference b/w Hashing and Encryption?

Ans Hashing :- Converts data into a fixed-length hash value, which is irreversible and used for integrity verification. (e.g. Password).

Encryption - Converts data into a cipher text that can be decrypted back to its original form using a key, ensuring confidentiality.

2. Define Message Authentication Code (MAC) and its importance in cryptography?

Ans A Message Authentication Code (MAC) is a short piece of information generated from a message and a secret key using a cryptographic algorithm. It is used to verify both the integrity and authenticity of a message.

Importance in Cryptography :- Ensure that the message is not altered and confirms the sender's identity, preventing tampering and forgery.

• Section - B ($2 \times 7 = 14$)

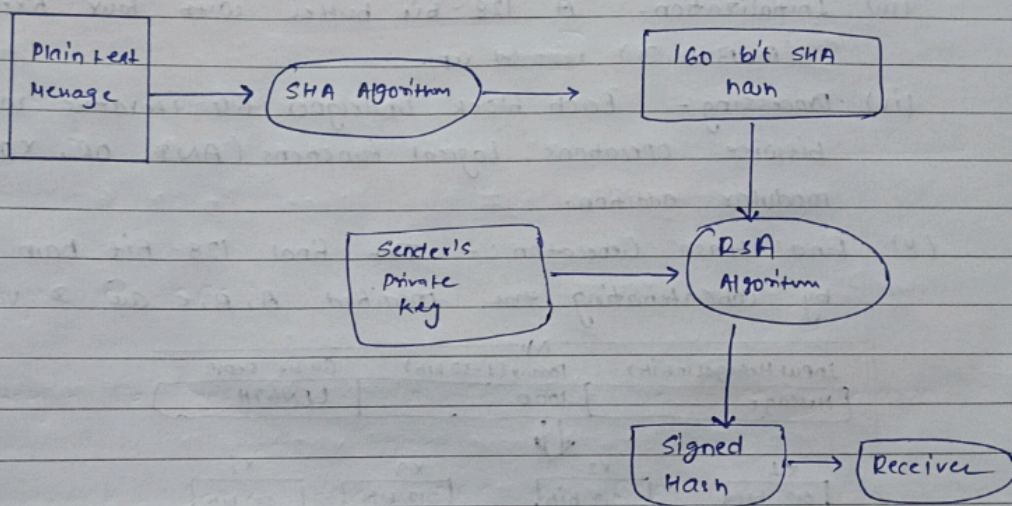
1. Describe the working of Secure Hash Algorithm (SHA). How is it used in digital security?

Ans SHA (Secure Hash Algorithm) is a cryptographic hashing algorithm that converts input data into a fixed-length hash value. It is widely used in digital security for data integrity, authentication and encryption.

• Working of SHA

(i) Input Processing - The input message is padded to ensure its length is a multiple of a predefined block size.

- (ii) **Message Block Division** - The padded message is divided into fixed-size blocks (e.g. 512-bit blocks for SHA-256).
- (iii) **Initialization** - A set of hash values (constants) is initialized.
- (iv) **Compression Function** :- Each block is processed through a series of mathematical operations, including bitwise shifts, logical functions, and modular addition.
- (v) **Final Hash Value** :- After processing all blocks, a final fixed-length hash is generated.



• **Uses in Digital Security** :-

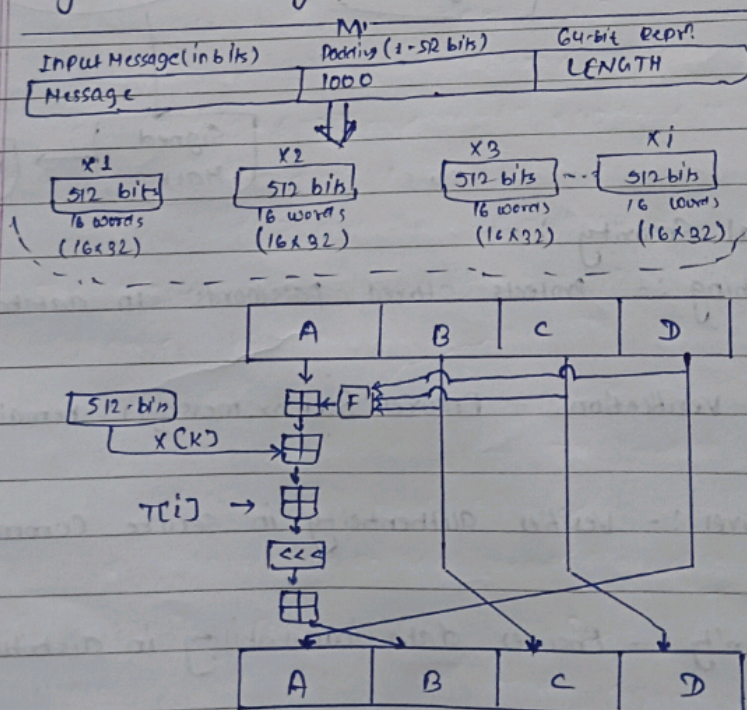
- (i) **Password Hashing** :- Protects stored passwords in databases.
- (ii) **Data Integrity Verification** :- Ensures files or messages remain unchanged.
- (iii) **Digital Signatures** :- Verifies authenticity in secure communications.
- (iv) **Blockchain Security** :- Ensures data immutability in distributed ledgers.
- (v) **SSL/TLS Encryption** :- Secures HTTPS communications.

Q2 Explain the working of Message Digest Algorithm (MD5) and its role in cryptography security?

Ans Message Digest Algorithm (MD5) is a cryptographic hash function that generates a 128-bit fixed length hash value from an input message.

Working of Message Digest Algorithm (MD5):

- (i) Padding - The input message is padded to make its length a multiple of 512 bits.
- (ii) Message Splitting - The padded message is divided into 512-bit blocks.
- (iii) Initialization - A 128-bit buffer with four fixed values (A, B, C, D) is set up.
- (iv) Processing - Each block undergoes 64 iterative rounds of bitwise operations, logical functions (AND, OR, XOR), and modular addition.
- (v) Final Hash Generation - The final 128-bit hash is obtained by concatenating the modified A, B, C and D values.



• Role in Cryptography Security :-

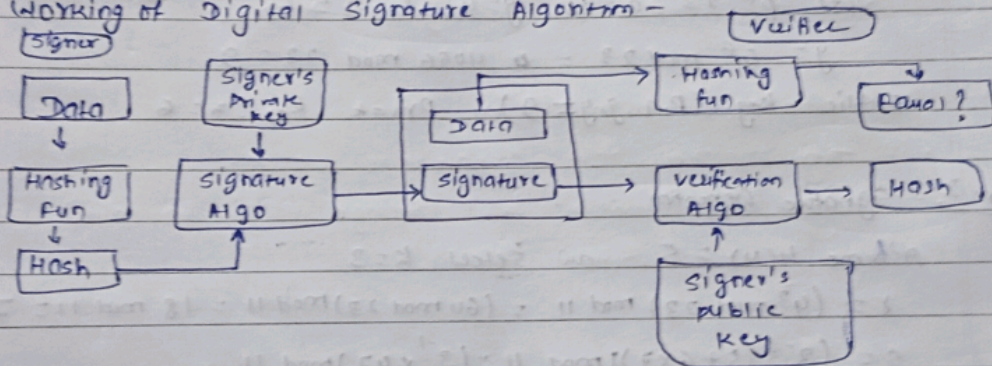
- (i) Data Integrity - Ensures file or messages are not altered during transmission.
- (ii) Password storage - Used for Hashing Passwords in older systems.
- (iii) Checksums - Verifies file integrity in downloads and software updates.
- (iv) Digital signature - Helps authenticate message and documents.

• Section-c (2x11=22)

1. State the working of Digital Signature Algorithm (DSA) with a suitable example?

Ans Digital signature Algorithm (DSA) is a cryptographic algorithm used for digital signatures to ensure data authenticity, integrity and non-repudiation.

Working of Digital Signature Algorithm -



Steps:

(i) Key Generation: -

- Choose a large prime no. p and a prime divisor q of $(p-1)$.
- Select a generator g such that $g = h^{(p-1)/q} \bmod p$.
- Choose a private key x , where $1 < x < q$.
- Compute the public key y as $y = g^x \bmod p$.

(ii) Signature Generation: -

- Select a random integer k , where $1 < k < q$.
- Compute $r = (g^k \bmod p) \bmod q$.

- Compute $s = K^{-1}(H(N) + z \cdot r) \bmod q$, where $H(N)$ is the hash of the message.
- The signature consists of the pair (r, s) .

(iii) Signature verification:-

- Compute $w = s^{-1} \bmod q$.
- Compute $u_1 = (H(N) \cdot w) \bmod q$.
- Compute $u_2 = (r \cdot w) \bmod q$.
- Compute $v = ((g^{u_1} \cdot y^{u_2}) \bmod p) \bmod q$.
- if $v = r$, the signature is valid, otherwise it is rejected.

Ex:- Step 1: Key Generate:-

Let $p = 23, a = 11, g = 4, z = 2 = 6$.

$$y = 4^6 \bmod 23 = 4096 \bmod 23 = 9$$

Public Key: $(P, a, g, y = 9)$, Private Key: $z = 6$.

Step 2: Signature Generation:-

Assume $H(M) = 5$, and select $K = 3$

$$r = (4^3 \bmod 23) \bmod 11 = (64 \bmod 23) \bmod 11 = 18 \bmod 11 = 7$$

$$s = (3^{-1}(5 + 6 \times 7)) \bmod 11 = (3^{-1} \times 47) \bmod 11$$

So, $3^{-1} \bmod 11 = 4$

$$\text{so } (4 \times 47) \bmod 11 = 188 \bmod 11 = 1.$$

The digital signature is $(r = 7, s = 1)$

Step 3: Signature Verification:-

$$w = 1^{-1} \bmod 11 = 1$$

$$u_1 = (5 \times 1) \bmod 11 = 5, u_2 = (7 \times 1) \bmod 11 = 7$$

$$v = ((4^5 \times 9^7) \bmod 23) \bmod 11$$

$$v = (1024 \times 4782969 \bmod 23) \bmod 11 = 7$$

So $v = 7$ and also $r = 7$

Since $v = r$ so the signature is valid.

2. Explain the concept of digital signature. How does it ensure authentication, integrity and non-repudiation? provide an example?

Ans A digital signature is a cryptographic mechanism that ensures the authenticity, integrity, and non-repudiation of digital messages or documents. It works like a handwritten signature but is more secure because it is mathematically generated using encryption techniques. Digital signatures use Public-Key cryptography (Asymmetric encryption), where a private key is used to sign the data, and a public key is used to verify it.

How digital ^{Signature} ~~Security~~ Ensure Security -

(i) Authentication :-

- Confirms the sender's identity by verifying the signature using their public key.
- Ensures the message was indeed sent by the claimed sender.

(ii) Integrity :-

- The signature is generated using a hash function (SHA-256, MD5, etc)
- Any change in the message alters the hash, making the signature invalid.

(iii) Non-Repudiation -

- Since only the sender's private key can generate the signature they cannot deny having signed the message.
- Prevents forgery or later denial of the transaction.

• example of digital signature process :-

Step 1: Signature Generation -

(i) Sender Ankit write a message :

- "Transfer 1000 rupees to Sumit"

- (ii) A hash function (SHA-256) generates a unique hash of the message.
- (iii) Ankit encrypts the hash with her private key ~~or~~ using RSA or DSA.
- (iv) The encrypted hash (digital signature) is attached to the message and sent to Sumit.

• Signature Verification: -

- (i) Sumit receives the message and signature.
- (ii) He generates a hash of the received message using the same hash function.
- (iii) Sumit decrypts the received signature using Ankit's Public Key to retrieve the original Hash.
- (iv) He compares both hashes: -
 - if they match, the signature is valid.
 - if they don't match, the message was tampered with or is not from Ankit.

End